

12 Annex, variant II: Activation of the lift/carousel via REST web service (only for instructed installation and maintenance personnel as well as IT specialists)



This chapter is intended for instructed installation and maintenance personnel and IT specialists only.

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1 Introduction

Using the REST-compliant "JRS" web service, various functions are possible on the MP control system, such as activating compartment LEDs, querying lift/carousel conditions or executing lift/carousel runs. The purpose of this web service is primarily machine-to-machine communication; it provides an alternative to the SOAP web service.

- ◆ Documentation is provided as an OpenAPI definition (see Chapter 2, page 134).
- ◆ The web service presents resources in the JSON format.
- ◆ The web service can be protected using the HTTP basic authentication process (see Chapter 4, page 138).
- ◆ Lift runs always have to be initiated using the **【↔】** key on the MP control system.
There is an exception only in conjunction with supplementary module "Remote-controlled Lift Operation with special safety equipment".

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2 Interface description

OpenAPI definition

The OpenAPI specification (OAS) defines a standardised, language-independent interface to HTTP APIs, which enables both humans as well as computers to discover and understand the skills of a service.

An OpenAPI definition may be used for the following purposes:

- For presenting documentation (e.g. Swagger Editor / UI)
- For programs for code generation
- For test programs

The OpenAPI definition can be downloaded via the MP control system. To do so, the following URL has to be entered in a Web browser connected to the MP control system:

http://<IP address of the MP control system>/jrs/openapi.<Format>

<Format>:

- yaml
- json

Depending on which format has been selected, the web browser used shows the OpenAPI definition (JSON) or starts to download it automatically (YAML).

Even though authentication is activated, the OpenAPI definition can be downloaded without entering login data.

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Excerpt from OpenAPI definition "openapi.yaml"

Configuration of OAS 3.0.1	Contents
Info object	<pre> info: title: Haanel REST API description: - Haanel REST API - HOST-WEB Protected intellectual property. All rights reserved by Hänel® GmbH & Co.KG Terms used: - '[LL]' = Lean-Lift - '[MS]' = Multi-Space - '[RO]' = Rotomat version: 0.1.0 </pre>
Server object	<pre> servers: - url: "{protocol}://mp14-xxxxxxx/jrs" variables: protocol: description: Application layer protocol default: http enum: - http - https </pre>
Tag object	<pre> tags: - name: Features description: Methods for supplementary features - name: Modules description: Methods for supplementary modules - name: Shelf description: Shelf methods - name: Standard description: Standard methods </pre>
Paths object	<pre> paths: /status: get: tags: - Standard summary: Query status information description: "With this method, the host has the ability to query various status\ \ information. \nThe method is non-blocking." operationId: hostweb/getStatus responses: "200": description: Successful operation content: application/json: schema: properties: code: type: integer format: int32 example: 200 data: \$ref: '#/components/schemas/ReadStatusRes' allOf: - \$ref: '#/components/schemas/ApiResponse' </pre>
Components object	<pre> components: schemas: ApiResponse: type: object properties: code: type: integer description: HTTP status code format: int32 status: type: string description: Status type of the response enum: - success - fail - error </pre>

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	<pre>message: type: string description: "A meaningful, end-user-readable message, explaining what went\ \ wrong \n(set if status = 'error')\" data: type: object externalDocs: description: Documentation about the message format url: https://github.com/omniti-labs/jsend ... securitySchemes: BasicAuth: type: http scheme: basic</pre>
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3 Overview of individual methods

Method		Description
GET /status		Query status information
PUT /shelves/{shelfId}		Get shelf with graphical display of the storage location and additional text
PUT /shelves/current	LL + MS	Store shelf
GET /shelves	LL + MS	Query of shelf numbers of all existing shelves
POST /shelves	LL + MS	Add shelf
DELETE /shelves/{shelfId}	LL + MS	Remove shelf
PUT /optimisation-run	LL + MS	Starting an optimisation run
PUT /location-indicator		Clear storage location display
PUT /peripherals/{deviceId}		Enable the interface for a peripheral device
PUT /base-services		Call up basic functions
GET /modules		Query supplementary modules
GET /features		Query supplementary features
PUT /start-url		Change the start-URL of the host Web server.
GET /start-url		Query start-URL of the host web server
GET /confirmation-key		Query confirmation key
Methods for supplementary features and supplementary modules		Description
PUT /sliding-door		Closing the automatic sliding door (Automatic sliding door), (Rotomat access code management)
PUT /signals/{signalId}		Activate signal elements of the signal column. (Signal column)
PUT /picking-displays/{displayId}		Write to and query requisition processing strip (requisition processing strip)
PUT /pre-positioning		Switch the shelf pre-positioning mode on/off (Shelf pre-positioning)
GET /shelf-props/heights	LL + MS	Retrieve shelf target height and stored article height (Storage location height management)
PUT /shelf-props/heights/{shelfId}	LL + MS	Change the shelf target height of a shelf (Storage location height management)
GET /shelf-props/speeds	LL + MS	Query shelf speed (Adjustable shelf speed)
PUT /shelf-buffer	LL	Get shelf in background to intermediate shelf buffer (Intermediate shelf buffer for job)
PUT /shelf-transfer	LL + MS	Start shelf transfer (Shelf transfer)

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4 Authentication

HTTP authentication mechanism

RFC 7235 defines an HTTP authentication mechanism which can be used by a server to trigger a client authentication request and by a client to provide authentication information.

The authentication process between the client and server functions as follows:

1. The server answers a client with the status code 401 (Unauthorized) and supplies information regarding the authorization using the response header field "WWW-Authenticate".
2. The client authenticates itself to the server by adding the header field "Authorization" to the request, which contains the required login information.
3. After a successful check of the login information, the request is processed by the server. If the check fails, the request is answered with status code 401 (Unauthorized).

The generic HTTP authentication mechanism forms the basis for a range of authentication processes.

Basic authentication process

The basic authentication process is defined in RFC 7617. Here, the login information is transferred as a user ID/password pair in Base64 coding.

If the server answers with status code 401 (Unauthorized), the "WWW-Authenticate" header field is set up as follows:

HTTP/1.1 401 Unauthorized
WWW-Authenticate: Basic realm="jrs", charset="UTF-8"

In order for verification to be successful, the client must fill in the "Authorization" header field with valid login information.

Scheme: Authorization: Basic base64(<user ID>:<password>)
Example: Authorization: Basic dXNlcjoyMWlzb25seWhhbGZ0aGV0cnV0aA==



Caution:

The basic authentication scheme that is used sends the login information in encoded form, not encrypted. For a secure exchange of login data, a connection using HTTPS/TLS is necessary.



For the definition of user and password, see Chapter 4.1, page 32 and on - "Settings - Host parameters" function

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Customer form annex for integration into corporate network

Ethernet connection to corporate network

This form is to be filled in by the customer's IT specialists if the service has been ordered by "Integrating a web application into the MP 14-H[HOST-WEB] SOAP for host communication via HTML pages using the HTTP protocol and lift/carousel activation using a web service via SOAP protocol".

Please fill in this form and give it to the Hänel Service technician when the MP control system is connected to the company network.



The "Technical Description of the Microprocessor Control System MP 14 Lean-Lift and Multi-Space" or "Technical Description of the Microprocessor Control System MP 14 Rotomat" is the basis for the following tables to be filled in:

MP 14-H-[HOST-WEB] - Connecting to a corporate network	
1.	Lift number: _____ (Lift/carousel to be integrated into the corporate network)
2.	Is the lift to receive the TCP/IP address via a DHCP server? ☐ Yes ☐ No
3.	x Fill in only if the answer to "2" was "No": Which TCP/IP parameters should the lift receive? TCP/IP address : _____ Sub-network mask : _____ Standard gateway : _____
4.	The lift uses another TCP/IP address internally for the MP multi-unit network. This is always active. It can be located in the B-class network in the address range 172.16.xxx.xxx to 172.31.xxx.xxx. Which address range is free: 172._____.xxx.xxx

Notes:

_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

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14.1 Overview of optional supplementary descriptions

In the following list, the optional Supplementary Descriptions that have been included in the documentation folder are ticked. The list provides no information about possible applications and combinations of supplementary modules and supplementary features.

	Lift/carousel version according to order confirmation Ausführung des Lifts gemäß Auftragsbestätigung	The marked (x) supplementary descriptions are part of the machine documentation Markierte (x) Zusatzbeschreibungen sind Teil der Maschinendokumentation
	Module 00) Remote-controlled Lift Operation (LL, MS, RO) Modul 00) Ferngesteuerter Liftlauf (LL, MS, RO)	Supplementary Description of the Remote-controlled Lift Operation Zusatzbeschreibung Ferngesteuerter Liftlauf MP-BEDIENZ-FERN.MP_14
	With add-on module 04) Storage location height management (LL, MS) Modul 04) Verwaltung der Lagerorthöhe (LL, MS)	Supplementary Description of the Storage Location Height Management Zusatzbeschreibung Verwaltung der Lagerorthöhe MP-BEDIENZ-HOEHE.MP_14
	With add-on module 24) Adjustable shelf speed (LL, MS) Modul 24) Einstellbare Tablargeschwindigkeit (LL, MS)	Supplementary Description of the Adjustable Shelf Speed Zusatzbeschreibung Einstellbare Tablargeschwindigkeit MP-BEDIENZ-GESCHW.MP_14
	With add-on module 25) Shelf pre-positioning by host (LL, MS, RO) Modul 25) Tablarvorpositionierung durch Host (LL, MS, RO)	Supplementary Description of the Shelf Pre-positioning Zusatzbeschreibung Tablarvorpositionierung MP-BEDIENZ-PREPOS.MP_14
	With add-on module 28) Intermediate shelf buffer for job processing (LL) Modul 28) Tablar Zwischenpuffer bei Auftrag (LL)	Supplementary Description of the Intermediate Shelf Buffer for job processing Zusatzbeschreibung Tablar Zwischenpuffer bei Auftrag MP-BEDIENZ-TABPUF.MP_14
	With add-on module 29) Shelf transfer (LL, MS) Modul 29) Tablartransport (LL, MS)	Supplementary Description of the Shelf Transfer Zusatzbeschreibung Tablartransport MP-BEDIENZ-TABTRA.MP_14
	Module 30) Extractor park position (LL, MS) Modul 30) Park-Position Extraktor (LL, MS)	Supplementary Description of Extractor Park Position Zusatzbeschreibung Park-Position Extraktor MP-BEDIENZ-PARKPOS.MP_14

LL = Lean-Lift
MS = Multi-Space
RO = Rotomat

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	Lift/carousel version according to order confirmation Ausführung des Lifts gemäß Auftragsbestätigung	The marked (x) supplementary descriptions are part of the machine documentation Markierte (x) Zusatzbeschreibungen sind Teil der Maschinendokumentation
	With electrically opening sliding door (RO) mit elektrisch öffnbarer Schiebetür (RO)	Supplementary Description of the Automatic Sliding Door Zusatzbeschreibung Automatische Schiebetüre MP-BEDIENZ-AUTOSD.MP_14
	With Automatically Lowering Front Edge of Shelf (RO) mit automatisch absenkbarer Tablarvorderkante (RO)	Supplementary Description of the Lowerable Front Edge Zusatzbeschreibung Absenkbare Vorderkante MP-BEDIENZ-AVK.MP_14
	(RO) (RO)	Supplementary Description of the Access Lock Zusatzbeschreibung Zugangssperre MP-BEDIENZ-CODE.MP_14
	High-speed door between access opening and drive shaft (LL, MS) Schnell-Lauftor zwischen Entnahmeöffnung und Fahrschacht (LL, MS)	Supplementary Description of the High-speed Door Zusatzbeschreibung Schnell-Lauftor MP-BEDIENZ-STOR.MP_14
	With automatically opening and closing sliding door (RO, LL) mit automatisch öffnende und schließende Schiebetüre (RO, LL)	Supplementary Description of the Automatic Sliding Door Zusatzbeschreibung Automatische Schiebetüre MP-BEDIENZ-AUTOSD.MP_14
	Automatic extraction of container (LL) automatischer Ein-/Ausschub eines Containers (LL)	Supplementary Description of the Automatic Shelf Ejection Zusatzbeschreibung Automatischer Tablarausschub MP-BEDIENZ-ATA.MP_14
	With container weighing device in the access point (LL, MS) mit Container-Wiegeeinrichtung in der Entnahmestelle (LL, MS)	Supplementary Description of the Shelf Weighing Device Zusatzbeschreibung Tablarwiegeeinrichtung MP-BEDIENZ-WEIGHT.MP_14
	Special Access Point for Operation with Standard / High Lift Truck and Container in Dimensions ..., in Special Version for Barrier On All Sides (LL) Sonderentnahmestelle für Bedienung mit Standard- / Hochhubwagen und Container in den Abmessungen ..., in Spezialausführung für umlaufende Begrenzung (LL)	Supplementary Description of the Automatic Shelf Locking for Operation with High Lift Truck or Forklift Truck Zusatzbeschreibung Automatische Tablarverriegelung für eine Bedienung mit Hochhubwagen oder mit Gabel-Hubwagen MP-BEDIENZ-TABVER.MP_14
	Unit prepared for containers with power socket (LL) Gerät vorbereitet für elektrifizierte Container (LL)	Supplementary Description of Shelves with Power Socket Zusatzbeschreibung Elektrifizierte Tablare MP-BEDIENZ-ELTAB.MP_14

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	Lift/carousel version according to order confirmation Ausführung des Lifts gemäß Auftragsbestätigung	The marked (x) supplementary descriptions are part of the machine documentation Markierte (x) Zusatzbeschreibungen sind Teil der Maschinendokumentation
	With camera holder and Hänel camera (LL, MS) mit Kamerahalterung und Hänel Kamera (LL, MS)	Supplementary Description of the Camera Zusatzbeschreibung Hänel-Kamera MP-BEDIENZ-CAM.MP_14
	With Pick-O-Light system, moving in the access point (LL, MS) mit Leuchtpunktanzeige, verfahren in der Entnahmestelle (LL, MS)	Supplementary Description of the Pick-O-Light system (variable) Zusatzbeschreibung Leuchtpunktanzeige (verfahrend) MP-BEDIENZ-LPAVER.MP_14
	Requisition processing strip (LL, MS, RO) Kommissionierleiste (LL, MS, RO)	Supplementary Description of the requisition processing strip Zusatzbeschreibung Kommissionierleiste MP-BEDIENZ-VFDLEISTE.MP_14
	With demo function (with trade fair demo function lift run warning sign) (LL, MS, RO) mit Messefunktion Demolauf (mit Warnschild Messe-Liftlauf) (LL, MS, RO)	Supplementary Description of the Demo Function Zusatzbeschreibung Messefunktion MP-BEDIENZ-DEMO.MP_14

LL = Lean-Lift
MS = Multi-Space
RO = Rotomat

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14.2 Revision notes

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- Complete revision starting from Chapter 9, extension of variant II: Activation of the lift/carousel via REST web service

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